Real-source retrieval. Traceable research leads.

HelioSense retrieval proof of concept

A fictional glucose-sensor company asks which Bern research capabilities could support its R&D. This experiment shows how the backend retrieves real institutional passages and makes the results inspectable.

Experiment: 19 October 2026, 09:22 UTC. Presentation revised 19 October 2026; original measurements retained.

Corpus capture	Queries	Passage occurrences	Documents
127,902 chunks	4	120	52

What worked

All 120 recorded exact-passage checks and 120 source-body hash checks passed. Retrieved publications include documented sensing, closed-loop systems and glucose-prediction research. The next page shows three examples.

Query	Encoding + search	Hits
full-demo-brief	24.824 s	30
glucose-sensors	21.839 s	30
cgm	18.119 s	30
biosensing-lay-terms	17.542 s	30

Method

Four requests used the full company brief and alternative glucose-sensor, CGM and everyday biosensing wording. Lexical and vector search ran independently; reciprocal rank fusion combined their results. The queries contained no researcher names. Examples were selected after inspecting the rankings.

What this establishes

Working retrieval and source traceability. This run returned no accepted structured entity profiles, so researcher mentions are leads extracted from publications. Expert Recall@10 was not measured. The broader application pipeline was tested separately (page 3).

Timing is one observation per query, including hosted encoding. It is the original baseline, not a measurement of the latest SQL optimization.

Selected research leads

Real publications support these examples. Publication dates identify their historical context; confirm current roles and collaboration availability before outreach.

Lilian Witthauer

Direct sensing research lead in a 2023 source

The retrieved interview identifies Witthauer with diabetes technology at UniBE/DCB and describes a continuous blood-glucose sensor. This supports an exploratory sensing lead for HelioSense.

> Lilian Witthauer ist Assistenzprofessorin für Diabetestechnologie an der Universität Bern und dem Diabetes Center Berne (DCB).
>
> Eines meiner aktuellen Projekte be - fasst sich mit der Entwicklung eines Sensors, der kontinuierlich und verzö- gerungsfrei den Glukosegehalt im Blut misst.

Original institutional source | biosensing-lay-terms, passage rank 3

José García-Tirado

Adjacent systems/control capability in a 2024 source

The DCB publication lists García-Tirado with closed-loop systems. The same passage describes glucose sensors, insulin pumps and algorithmic control. This is a documented systems connection, not proof of a HelioSense-compatible sensor.

> Prof. Dr José García-Tirado, Systèmes en boucle fermée
>
> technologiques composés de capteurs de glucose et de pompes à insuline,

Original institutional source | biosensing-lay-terms, passage rank 9

Stavroula Mougiakakou

Historical diabetes algorithms lead; follow-up verification needed

The ARTORG report names Mougiakakou in Diabetes Technology Research and discusses glucose-profile prediction with statistical/computational methods. This supports historical relevance to signal processing.

> Diabetes Technology Research Stavroula Mougiakakou, Technical Head of Research Group
>
> Data-driven models, based on advanced statistical and computational intelligence meth- ods are developed for the prediction of glucose profile in the near future

Original institutional source | glucose-sensors, passage rank 19

One ranking weakness

A rehabilitation page ranked first for the sensor query. Retrieved passages therefore still need a relevance assessment. Passage rank is not a validated researcher score.

The JSON retains every ranking, selected quotation, source URL, character offset and body hash. Whitespace in displayed quotations is normalized for readability.

Evidence, scope and reproduction

Three separate checks

Evidence	Result and scope
This retrieval experiment	Real corpus; fictional company. Passage integrity and selected research leads. No reranking, web research or Apertus drafting in this run.
Later pipeline acceptance	Actual Apertus with fictional sources and local PostgreSQL checkpoints: match, clarification/resume, assessed report, review and PDF export. Separate restart tests used scripted providers.
Later live SQL measurements	Six-input retrieval over 144,268 chunks completed in 6.04-7.43 s; complete RRF returned 30 hits in 6.81 s with cached real query vectors. Encoding and reranking excluded.

The two timing sets use different captures and include different work; they are not a like-for-like speedup. Remaining acceptance is expert recall/coverage, a fresh complete case on the optimized live corpus, and the selected deployment environment.

Inspect or rebuild

The committed receipt and this PDF can be inspected without services or credentials. Rebuild the presentation from repository root with the installed backend:

```
python -m backend.evaluation.render_proof_of_concept
```

The renderer reads docs/proof-of-concept/retrieval-results.json and writes only the PDF. It checks receipt consistency; it does not rerun retrieval or revalidate private source bodies.

To collect a new experiment, use an existing authorized run and matching encoder settings:

```
python -m backend.evaluation.proof_of_concept --env-file PRIVATE_ENV --run-id OWNED_RUN_ID
```

This calls the configured encoder and database. Current retrieval permits temporary tables, with no persistent corpus/case writes. A new run produces a separate result, not the original measurement.

Recorded provenance

Field	Original experiment
Schema	0012_membership_lookup
Snapshot	658224cd-9e05-4fa1-bbbb-bc7d309f7fdf
Source	a35d70af7f65 + local changes
Encoder	google/embeddinggemma-2; 768 dimensions
Encoder revision	914f7f89142e33e77833254d9c9b90c3cef7303b

Receipt SHA-256: 2638c5f5dc28fc37891c0dc0d768386aef90b07cf8c4f2eff3290fb24ca58f0f

Related records: docs/proof-of-concept/README.md; docs/project/PIPELINE_RELIABILITY.md; docs/project/FRONTEND_RECOVERY.md; docs/data/LIVE_RETRIEVAL_PERFORMANCE.md.